

Bayu Rizky Kurnia Pratama

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SUMMARY

I am a Backend Engineer and System Design Enthusiast with practical experience in building scalable applications. I have developed robust backend systems, including virtual economies and gamification features, utilizing Node.js and PHP. Demonstrating strong collaborative skills, I have coordinated cross-functional teams to deliver end-to-end solutions across web and mobile platforms. I am passionate about system infrastructure and integrating advanced technologies like LLMs to build impactful, forward-thinking products.

EXPERIENCE

Fullstack Developer

Jul 2026 – Present

Kaimify | Remote

- Developed the backend of Kaimify, a wellness super-app, using Flutter, NestJS, PostgreSQL, and Drizzle ORM, with automated CI/CD deployment on a self-managed VPS.
- Developed and integrated Flutter features with REST APIs, collaborating across the mobile and backend stack to deliver cross-platform functionality.
- Built a dual-currency wallet system with ACID-compliant transactions, integrated with Google AdMob SSV reward validation and in-app purchase subscription webhooks.
- Integrated LLM services into the journal module to analyze user entries and generate personalized insights and feedback.
- Developed core application modules, including Marketplace, Daily Missions, Streaks, Achievements, and a unified Calendar API that aggregates habits, tasks, journals, and the Balinese calendar (Wariga and Otonan).
- Applied an event-driven architecture and scheduled background jobs to handle asynchronous workflows and recurring system tasks.
- Improved database performance by eliminating N+1 query patterns with batch fetching and in-memory indexing, and implemented cursor-based pagination for wallet transaction endpoints.

Programming Teacher

Aug 2024 – Present

Timedoor Indonesia | Bali

- Delivered bilingual (English and Bahasa Indonesia) programming instruction to a diverse student base, ranging from children to adult professionals.
- Facilitated foundational logic and game development classes for kids and teens utilizing Scratch, Roblox Studio, and Construct.
- Instructed adult learners (Pro class) in advanced full-stack web development (HTML, CSS, JavaScript, Vue, Laravel) and data science fundamentals (Python, Machine Learning, and AI).

Undergraduate Researcher

Jan 2026 – Jun 2026

Point of Sale Project | Udayana University

- Architected a multi-tenant and offline-first Point of Sale application utilizing Clean Architecture principles to ensure high system resiliency for SMEs.
- Engineered the Data Layer to map local SQLite databases with cloud-based Supabase and implemented bidirectional synchronization queues using the Brick library.
- Resolved data conflicts through Event Sourcing and a Last Write Wins timestamp strategy while enforcing strict data privacy isolation via Row Level Security (RLS).
- Designed the Domain Layer to separate business logic from physical storage details and utilized middleware to manage access hierarchies directly at the Use Case level.
- Formulated comprehensive integration testing with over 300 scenarios to ensure reliable business logic execution and data consistency.
- Validated system resiliency and offline-first capabilities by successfully synchronizing 1,000+ simulated offline transactions with zero data loss upon network reconnection.
- Optimized local data processing, maintaining sub-20ms latency during transaction execution under simulated poor network conditions.

Frontend Developer Intern

Mar 2025 – Jun 2025

Timedoor Indonesia | Bali

- Developed and maintained scalable web interfaces using Vue.js.
- Implemented REST API integration to manage dynamic data and frontend business logic.
- Resolved 100+ frontend bugs across low to high severity levels, including logical errors and UI issues.
- Enhanced application stability, performance, and usability through systematic debugging and refactoring.

- Collaborated with backend engineers to ensure accurate data flow and API compatibility.
- Applied Git version control for efficient collaboration and code management.

Cloud Computing Cohort
Bangkit Academy | Remote

Sep 2024 – Jan 2025

- Graduated with Distinction, ranking in the top 10% of my Cloud Computing path with consistent performance.
- Developed a capstone project, Memotions, a journaling app with machine learning and gamification features, addressing mental health challenges. The project ranked in the Top 50 out of 644 submitted projects.
- Built REST APIs using Express.js and TypeScript, deploying on Google Cloud Platform (GCP) with Cloud Run and integrating Firebase Cloud Messaging for notifications.
- Utilized PostgreSQL and Drizzle for database management, with secure authentication via JWT.
- Led a diverse team of 7 engineers as Team Manager, ensuring collaboration across backend, machine learning, and Android development roles.
- Recognized as one of the Top 5 Best Presenters in the English ILT Activity Class.

Vice Chairman
Technology Artisan | Jimbaran, Bali

Jan 2024 – Dec 2024

- Planned programs related to scientific fields and technological advancements.
- Coordinated with 6 divisions: Software Engineering, Data Engineering, Cyber Security, UI/UX, Smart Technology, and Business with Technology.
- Interacted with external parties, such as speakers, mentors, and professionals in relevant fields.

Teaching Assistant
Udayana University | Jimbaran, Bali

Sep 2023 – Jun 2024

- Served as Head Teaching Assistant for Algorithm and Programming for 1st-semester students, coordinating assistant teams and collaborating with lecturers to develop C/C++ practicum modules.
- Assisted 2nd-semester students in Database Concepts practicum, helping design modules covering MySQL relational databases, Entity Relationship Diagrams (ERD), and Physical Data Models (PDM).
- Supervised and guided undergraduate students during regular laboratory sessions to resolve programming issues and implement relational database schemas.

EDUCATION

Udayana University
Bachelor of Computer Science
Major in Information Technology
Cumulative GPA: 3.93/4.0
Relevant Coursework: Algorithm and Programming, Database Concept, Computer Network, OOP, UI/UX, Linear Algebra, Data Warehouse, Internet Programming, Software Engineering

Aug 2022 – Aug 2026

AWARDS & HONORS

1st Place of ICP Indonesia Hackathon
ICP Indonesia Hackathon | Ash Nuanu, Tabanan

May 2025

- Won First Place at the ICP Indonesia Hackathon as part of team Hackathon Dadakan.
- Developed Moodyan, a decentralized journaling application focused on data privacy and mental health.
- Built a fully functional Web3 application using React, TypeScript, Internet Computer Protocol (ICP), and Motoko.
- Implemented blockchain-based authentication using Internet Identity and AI-powered mood analysis with Llama 3.2:8B.
- Successfully delivered an MVP despite first-time exposure to Motoko and Web3 technologies.

3rd Place of IT Business Development Competition
PNB IT Competition #16 x ECO | Politeknik Negeri Bali

Oct 2024

- Awarded 3rd place for developing TemanTaman, a digital auction platform for ornamental plants, addressing limited market access, high pricing, and restricted reach for plant sellers.
- Validated market demand with 871,027,342 ornamental plants produced in 2023, reflecting a 12% year-over-year growth.

- Designed an efficient auction-based transaction system where sellers list plants, buyers place bids, and the highest offer is accepted before secure delivery is arranged.
- Competed under the theme: “Millennials’ Innovation in Renewable Technology as a Pillar for Building the Digital Era with Full Environmental Awareness.”
- Organized by Bali State Polytechnic, held offline from October 11–13, 2024.

Awardee of Program Beasiswa Cendekia BAZNAS
BAZNAS (Badan Amil Zakat Nasional) | Universitas Udayana

Aug 2024 - Aug 2026

- Selected for a prestigious national scholarship program focused on academic excellence, character building, and social leadership.
- Selected as a contributor for the BCB national publication, submitting a manuscript titled "Menyusun Hari, Tim, dan Baris Kode" in February 2026 that highlights the intersection of software engineering, technical leadership, and social impact.

VOLUNTEER

Head of Committee
Pesantren 1000 Cahaya Ramadan | Kuta, Bali

Feb 2025 – Mar 2025

- Spearheaded a large scale social and educational event targeting marginalized groups, including orphans and children with disabilities, as part of a national BAZNAS initiative.
- Innovated traditional learning methods by integrating digital gamification utilizing the Bloocket platform to significantly increase participant engagement.
- Managed cross functional coordination, logistical planning, and event execution while successfully navigating operational challenges such as limited data infrastructure and difficult location access.
- Authored the comprehensive accountability report, detailing program execution, budget administration, and strategic recommendations for future improvements.

Head of Scientific Committee
Pelatihan PKM-TI 2024 | Jimbaran, Bali

Jan 2024 – Jun 2024

- Managed members of the scientific committee regarding our tasks, including team and PKM proposal recapitulation, recording feedback and suggestions from supervising lecturers, and ensuring proposal format compliance
- Created rules and coordinated with supervising lecturers regarding those rules.
- Acted as project manager and web programmer for PKM (Program Kreativitas Mahasiswa) team and proposal management website within the student of the Information Technology department at Udayana University.

Head of Competitive Programming Committee
Information Technolgy Creative Competition 2023 | Jimbaran, Bali

Jun 2023 – Nov 2023

- Contributed to the planning and execution of national competitive programming competitions hosting up to 50 contestants.
- Managed members of the competitive programming committee regarding our tasks, including problem creation, preparation of the preliminary platform, and overseeing the competition execution.
- Coordinated with supervising lecturers regarding rules and competition programming problems.

ADDITIONAL

Spoken Languages: Bahasa Indonesia (Native) and English (Intermediate)
Technical Skills: TypeScript, JavaScript, PHP, Dart, SQL, C/C++, Python, Go, Node.js (Express, NestJS), Laravel, React, Next, Vue, Nuxt, Flutter, PostgreSQL, SQLite, Supabase, Git, GCP, REST API, and Web3 (Motoko, ICP)